Chemo Comfort: AI-Based Wearable for Early Detection of Chemotherapy Induced Peripheral Neuropathy

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BONAFIDE CERTIFICATE

Certified that this project report entitled "**Chemo Comfort: AI-Based Wearable for Early Detection of Chemotherapy Induced Peripheral Neuropathy"** is a bonafide work of **R. Gopikasree (23BEC1013) and Kirthana.S (23BEC1412) who** carried out the Project work under my supervision and guidance for **BECE403E-EMBEDDED SYSTEM DESIGN.**

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ABSTRACT

Chemotherapy-induced peripheral neuropathy is a common and debilitating side effect of several anticancer treatments, leading to long-lasting sensory, motor, and autonomic dysfunctions. One of the major challenges in the early detection of CIPN is caused by the dependence on subjective clinical scales and inconsistent input from patients. The following work presents Chemo Comfort: a compact, cost-effective, and intelligent diagnostic device designed for the early, objective detection of CIPN. It implements three fundamental modules: (1) a questionnaire interface on an OLED display for recording patient-reported outcomes based on a simplified CIPN-20 model; (2) a vibration perception test using a haptic actuator, DRV2605L, to assess large-fibre sensory function and reaction time; and lastly, (3) a thermal perception test using a Peltier module and temperature sensor to evaluate the small-fibre response to controlled cold and heat stimuli. The ESP32 microcontroller processes the data coming from all the modules and executes an Edge Machine Learning model that was previously trained on real-world patient data. The trained model (which includes conversion to TinyML, TensorFlow Lite format) was deployed directly on the ESP32 for on-device inference, classifying the severity of CIPN as low, medium, or high risk. This approach provides an opportunity for instant assessment without any cloud dependency, taking care of considerations such as privacy, portability, and latency. The resultant information will be displayed for immediate clinical interpretation on the OLED interface. Chemo Comfort can, therefore, offer a scalable solution for home-based or outpatient monitoring of CIPN by fusing a set of low-cost sensors with real embedded intelligence, which makes early intervention possible and optimizes the quality of life in patients.

TABLE OF CONTENTS

1. INTRODUCTION

1.1 MOTIVATION

CIPN is a common and debilitating side effect of many widely used chemotherapeutic agents, including taxanes, platinum compounds, and vinca alkaloids. This neuropathy presents itself as a debilitating spectrum of sensory, motor, and autonomic dysfunctions, such as pain, tingling, and numbness in the hands and feet. These symptoms not only significantly impair the patient's quality of life but also present a critical clinical challenge because severe cases may necessitate dose reduction or the complete cessation of therapy, thus potentially compromising treatment outcomes.

The clinical assessment for CIPN is presently insufficient. Standard protocols are based on self-reported questionnaires, such as the EORTC QLQ-CIPN20, and clinician-based grading systems like CTCAE. Such methods are inherently subjective, susceptible to inconsistency, and lack the sensitivity that could determine very small, early changes in nerve function. By the time neuropathy using these methods is diagnosed, significant damage may already be irreversible. Whereas objective "gold standards" like NCS or QST are accurate, they are expensive, non-portable, and confined to specialized clinics, hence completely impractical for routine or home-based monitoring.

\The large gap in available access to objective, low-cost assessment methodologies has motivated the design of Chemo Comfort: a portable embedded diagnostic system that uses multimodal sensory testing with intelligence on device. The system combines quantitative vibration and thermal sensory threshold evaluations with patient-reported outcomes. Data is analysed on-device by a TinyML model on an ESP32 microcontroller. Its aim is to yield an accurate, affordable CIPN screening tool in real time that would be ideal for clinical and remote settings because it enables longitudinal tracking and early detection while at the patient's home.

1.2 OBJECTIVES

The main objectives of this project are:

- The design and development of the hardware platform for performing vibration and thermal sensory threshold testing require compact, low-cost components like DRV2605L haptic drivers, Peltier modules, and digital temperature sensors.
- Integrate the patient-reported outcome data from the standardized EORTC QLQ-CIPN20 questionnaire directly into the interface device for a comprehensive assessment that includes subjective and objective parameters.
- Embed a clinically driven Machine Learning model, trained with clinically inspired datasets from open published studies such as PMC12324740, PMC5541789, and PubMed 33507480, and enhanced with synthetically generated data augmentations capable of CIPN risk level classification on an ESP32 offline.
- Validation and demonstration of edge inference capability by converting the trained TensorFlow model into TensorFlow Lite and deploying it on a microcontroller will be done to provide real-time decision-making without cloud dependency.
- The design will ensure portability, affordability, and usability by developing a system with low power consumption, not requiring any external computer, and presenting the results directly on an OLED interface for clinicians or patients to interpret.

1.3 APPLICATIONS

The developed ChemoComfort device has wide-ranging clinical and research applications due to its portable, objective, and low-cost design.

A. **Clinical Settings in Oncology:** The device allows oncologists and nurses to conduct rapid, quantitative chairside assessment of neuropathy prior to every cycle of chemotherapy. It offers immediate, objective data to supplement subjective patient reports, thus empowering clinicians to make appropriate, timely dose adjustments. Since neuropathy can be detected early in its subclinical stages by this proactive monitoring, the progression to severe and irreversible nerve damage may be averted.
B. **Home Monitoring and Telehealth:** The system is especially suitable for remote patient monitoring, making it possible for those on long-term chemotherapy to self-monitor neuropathy progression from home. This makes it possible to capture real-world changes longitudinally. This device can also transmit this data to health providers for telemonitoring to decrease the travel burden on patients and allow for early intervention as soon as sensory thresholds start to decline.
C. **Rehabilitation Centres**: It is a system that helps physiotherapists track sensory recovery objectively in patients with established neuropathies, both chemotherapy-induced neuropathy and diabetic neuropathy. The therapists can use this quantitative feedback to design, tailor, and then validate individual rehabilitation protocols, such as sensory re-education and balance training.
D. **Educational and Research Tool:** As a tangible platform that integrates biomedical sensing, embedded systems, and TinyML, the device stands out as an excellent tool for teaching advanced engineering and data science. It offers an affordable, adaptive system to researchers for further investigation into the pathophysiology of peripheral neuropathies and capturing new large-scale datasets.

2. EMBEDDED SYSTEM DESIGN PROCESS

2.1 REQUIREMENTS

A. Functional Requirements

The ChemoComfort system shall be a portable, low-cost system for the detection and classification of the severity of Chemotherapy-Induced Peripheral Neuropathy. It does this by combining patient-reported data with quantitative sensory tests. The functional goals are:

- Present the 20-item EORTC QLQ-CIPN20 questionnaire dynamically via an OLED display, while recording patient responses reliably thanks to intuitive input buttons.
- Therefore, the objective of this study was to quantify the vibration perception threshold using a haptic feedback motor-DRV2605L for precise assessment of large sensory fibre function.
- To carry out controlled thermal sensitivity testing, both warm and cold perception, by using a Peltier element with a digital temperature sensor DS18B20/TMP117 for accurate detection of small fibre neuropathy.
- To process the combined vibration, thermal, and questionnaire data through an embedded, on-device TinyML model to classify CIPN in real-time.
- To display the final predicted CIPN status, such as “No Neuropathy,” “Low,” “Moderate,” or “Severe” on the OLED screen for immediate interpretation by the patient or clinician.

B. Non-Functional Requirements

- Low Cost: Use widely available off-the-shelf components-ESP32, DRV2605L, and Peltier-to keep the overall bill of material cost well below ₹1,000-₹2,000 to make it very affordable for scalable deployment.
- Low Power Consumption: The system must be optimized for low-power operation, capable of being powered via a standard USB port or a portable battery pack.
- Compactness and Portability: The entire setup, which would include sensors and the main controller, needs to fit in an ergonomic, handheld, or small desktop module for easy transport and usage.

- Easy to Use Interface: Utilize a straightforward, button-activated interface displaying clear on-screen prompts. Minimal training should be needed for both the patient and clinician.
- Edge Intelligence (Privacy): All data processing and classification have to happen in the device itself. No cloud/internet connectivity needed ensures patient data privacy and functionality in offline settings.
- Reliability: For clinical viability, the system must provide consistent and repeatable measurement of sensory thresholds across multiple test sessions.

2.2 SPECIFICATIONS

Parameter	Specification
Microcontroller	ESP32-WROOM with dual-core Xtensa LX6 (240 MHz, 520 KB SRAM)
Power Supply	5V USB or Li-ion battery with MT3608 boost converter
Display	0.96” OLED (I²C, 128×64 pixels)
Vibration Module	DRV2605L Haptic Driver (LRA/ERM motor)
Thermal Module	TEC1-12706 Peltier module + DS18B20/TMP117 temperature sensor
Machine Learning Model	2-layer MLP (Dense 16 + Sigmoid Output), deployed via TensorFlow Lite Micro
Input Interface	5 tactile pushbuttons (4 for questionnaire options, 1 for response/reset)
Output	OLED display and optional serial monitor logging
ML Features	Vibration Threshold, Cold Threshold, CIPN20 Sensory, Motor, Autonomic Scores
Output Classes	Normal, Moderate , Severe
Accuracy (offline)	AUC ≈ 0.7 on test data
Response Time	<1 second inference latency on ESP32

2.3 ARCHITECTURE: HARDWARE AND SOFTWARE

1. Hardware Architecture

The system is designed to be an embedded, modular platform. It consists of the following subsystems connected:

- **Input Layer**

1. Questionnaire Interface: A set of tactile pushbuttons as GPIO inputs for navigation within the UI and patient response recording, such as "Yes," "No," "Next".
2. Vibration Sensing Unit: A DRV2605L haptic driver and an eccentric rotating mass (ERM) motor generate precisely controlled vibration intensities. The patient response is captured through a button press, for example, "I feel it".
3. Thermal sensing unit: A Peltier module is energized via a MOSFET H-bridge for bidirectional (hot/cold) control. A high-precision digital sensor measures the exact surface temperature of the fingertip (DS18B20/TMP117).

- **Processing Unit**

1. ESP32 Microcontroller: This is the core that utilizes a dual-core processor to handle all parallel tasks of signal generation (PWM for haptics/thermal), user interface control (OLED/buttons), peripheral data acquisition, and execution of local ML inference.
2. MT3608 Boost Converter: A dedicated step-up converter supplies a steady, higher voltage, needed by power-consuming thermal and vibration modules, regardless of the main logic level.

- **Output Layer**

1. OLED Display: A high-contrast I2C display shows all user-facing information, including instructions, questionnaire items, and the final diagnostic output.
2. Serial Debug Port (UART): This is used during development to log raw sensor readings and inference results in real-time for calibration, testing, and validation.

2. Software Architecture

Firmware is written in C++/Arduino, and is organized into the following layers:

- **Data Acquisition Layer**
  The main control loop reads button inputs for the UI and polls sensor data during testing procedures: temperature readings and current amplitude of vibration.
- **Feature Extraction & Normalization**

1. Extracts the key features from the tests: the VPT-the vibration amplitude at which the user responds-and the cold/warm temperature thresholds.
2. This normalizes the raw feature vector on-device using pre-computed scaling parameters (mean/std) derived from the original Colab model development.

- **TinyML Inference**

1. Layer Loads the quantized TFLite model compiled in `model.h` and executes a forward inference on the normalized feature vector using the `TFLite-Micro` interpreter.
2. This layer outputs the raw predicted CIPN probability, which is a float value between 0 and 1.

- **Display & Decision Layer**
  Transforms the raw probability into a human-readable qualitative risk category, such as "Low Risk," "Moderate," "Severe.
  It displays this final result on the OLED display and moves the device into a ready state to wait for the commencement of the next test session.

2.4 COMPONENTS: HARDWARE & SOFTWARE

- **Hardware Components**

Component	Description / Function
ESP32-WROOM	Core processing and ML inference unit.
DRV2605L Haptic Driver (×2)	Generates precise vibration stimuli for vibration threshold test.
Peltier Module (TEC1-12706)	Provides controlled cooling/heating for thermal testing.
DS18B20 / TMP117	Digital temperature sensor for feedback control of Peltier surface.
MOSFET H-Bridge	Switches current direction through Peltier element for heating/cooling.
OLED Display (0.96")	Displays questions, progress, and inference result.
Push Buttons (5×)	Four for questionnaire responses, one for reset/acknowledgement.

- **Software Components**

Software Tool / Library	Purpose
Arduino IDE	Embedded firmware development.
Adafruit_GFX & SSD1306 Libraries	OLED control and UI display.
Adafruit_DRV2605 Library	Haptic motor control.
TensorFlow Lite for Microcontrollers (tflm_esp32)	On-device inference engine.
EloquentTinyML	Simplified wrapper for deploying and running TFLite models on ESP32.
Python (TensorFlow + Sklearn)	Dataset preprocessing, model training, and conversion.
Pandas / NumPy	Dataset manipulation and feature scaling.

2.5 SYSTEM INTEGRATION

The integration of the system was done in three clear, successive steps: to ensure strong functionality and reliable performance.

- **Hardware Integration:** The primary focus in this first stage was the physical integration of the prototype and electrical verification. Each module (OLED display, vibration motor, thermal unit, temperature sensors, and input buttons) was separately tested with the ESP32 on a breadboard to verify basic functionality and validate pin mappings before being integrated into a single prototype. Extra attention was given to the power delivery subsystem: voltage levels had to be carefully verified using the MT3608 boost converter to ensure a stable, clean power supply, especially to the high-current Peltier and DRV2605L modules to avoid possible brownouts or inconsistent sensor readings at peak operation.
- **Software Integration:** Firmware was developed in a staged, modular fashion. The base code for the user interface, including the questionnaire logic and display menus, was created. Next, specific control functions for the sensory tests were written and unit-tested, such as ramp-up algorithms for vibration and thermal control. Lastly, integration of the TinyML inference engine was done. A header file, which is a C array generated from the Google Colab training pipeline, was included directly in the Arduino sketch, thus loading the pre-trained neural network through the TensorFlow Lite Micro interpreter for completely self-contained real-time prediction.
- **System Testing and Calibration:** Hardware combined with software; the entire system was put to a series of tests and calibrations. It was at this point that the sensory modules were benchmarked against known benchmarks for clinical relevance. For instance, the vibration and thermal thresholds were correlated with reference values derived from published clinical datasets; for example, the median VPT for CIPN patients versus healthy subjects was 17.90 and 7.73, respectively. Synthetic data was fed into the system to validate its predictive accuracy, and its outputs were compared to the predictions from the original Python model in order to confirm model parity. Once the device had been validated, it successfully achieved its goal of real-time classification, with the total inference latency consistently below one second.

3. SYSTEM IMPLEMENTATION

3.1 SOFTWARE

The main reasons for this division of tasks are to split the complete software implementation of the ChemoComfort system into two distinguished, yet complementary, environments because the resource requirements for the development and training of machine learning models are high, while the constraints are in the target embedded device. It consisted of the machine learning development and training environment, Python/Colab, and the embedded firmware environment for device-level execution in Arduino IDE.

Machine Learning Environment

All model development was performed in a Google Colab notebook (Python 3.10, TensorFlow 2.2). Due to the lack of a public dataset combining quantitative sensory data with CIPN20 scores, a realistic dataset was synthesized, drawing upon statistical patterns from key research papers.

The synthesized dataset included five key input features: VPT, cold detection temperature, and the CIPN20 Sensory, Motor, and Autonomic scores.

A two-layer MLP model was chosen, featuring one hidden dense layer with 16 ReLU activation units and a final sigmoid output neuron for binary CIPN classification. The scikit-learn and TensorFlow pipeline handled missing values via median imputation (SimpleImputer) and normalized features with StandardScaler. The dataset was split 85:15 (train:test) and the model was trained for 30 epochs using the Adam optimizer (learning rate = 0.001), with performance evaluated by the Area Under the Curve (AUC) metric.

Following training, the model was converted to the TensorFlow Lite format. It was then further optimized using post-training INT8 quantization to reduce its memory footprint for the ESP32. A custom Python script automatically converted the resulting tflite model file into a C header file (model.h), containing the model as a large byte array ready for compilation.

```
=== Final summary ===
Keras test AUC           : 0.7613
Keras test accuracy      : 0.7000
Keras precision/recall/f1 : 0.6042/0.4874/0.5395
Float TFLite file        : model.tflite  size=2140 bytes (2.1 KB)
Quant TFLite file        : model_quant.tflite  size=2512 bytes (2.5 KB)
```

Fig. 3.1: Model accuracy and tflite size

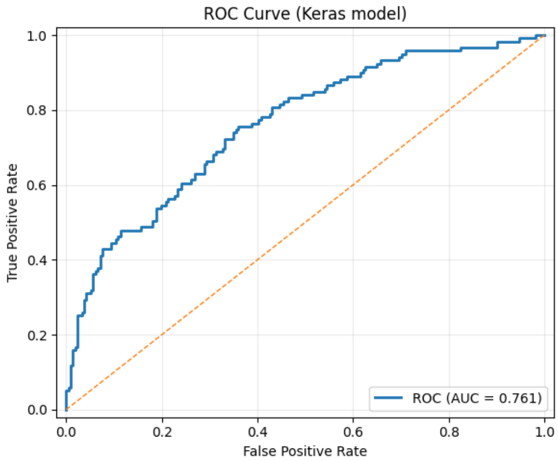

Fig. 3.2: ROC curve plot

Embedded Firmware Environment

The embedded firmware that runs directly on ESP32 was developed using the Arduino IDE in C++. This environment is chosen for its excellent library support, ease of use for rapid prototyping, and strong community backing for the ESP32 platform. Key libraries that formed the software stack include:

- **Adafruit_GFX and Adafruit_SSD1306**: Used together to drive the I2C OLED display; they handle everything from basic text and shape drawing to the multi-page questionnaire interface management.
- **Adafruit_DRV2605**: The library 'Adafruit_DRV2605' encapsulates the operations required to control the haptic motor to simply write straightforward values instead of manage complicated PWM signals which provide for fine settings of vibration strength.
- **tflm_esp32 and eloquent_tinyml**: These are the core libraries for on-device intelligence, providing functions to load the TensorFlow Lite model, allocate memory (the tensor arena), and run the inference.
- **Wire.h:** This library implements the I2C communication protocol, which underpins how the ESP32 communicates with the OLED display, DRV2605, and thermal sensor.

The main code, implements the entire user workflow. Upon start-up, it:

- Initializes all hardware: display and sensors.
- This displays the 20-question CIPN20 survey on the OLED, guiding the user through each question and recording their input via the tactile buttons.
- Starts the VPT test, which gradually increases the amplitude of the vibration using the DRV2605L and records the exact value at which the user can feel the sensation through a button press.
- Performs the thermal test by using the Peltier element and temperature sensor, cooling the surface until the user reports sensing the cold, at which point the cold perception threshold temperature is recorded.
- Normalizes all these collected parameters-questionnaire scores, VPT, and thermal threshold-using the same scaling factors as the training environment.
- Feeds this normalised feature vector to the embedded TFLite model for on-device inference.
- Finally, the device displays the resulting probability of CIPN and its qualitative counterpart, such as "Moderate Risk," on its OLED screen.

Everything, from data collection to final result, was optimized for a clinical setting. The model's on-device inference time was consistently measured at less than one second and thus provided instant feedback. Crucially, the predictions made by the embedded model were validated to be very near those of the original desktop Python model, AUC ≈ 0.7, confirming that deployment was successful and accurate.

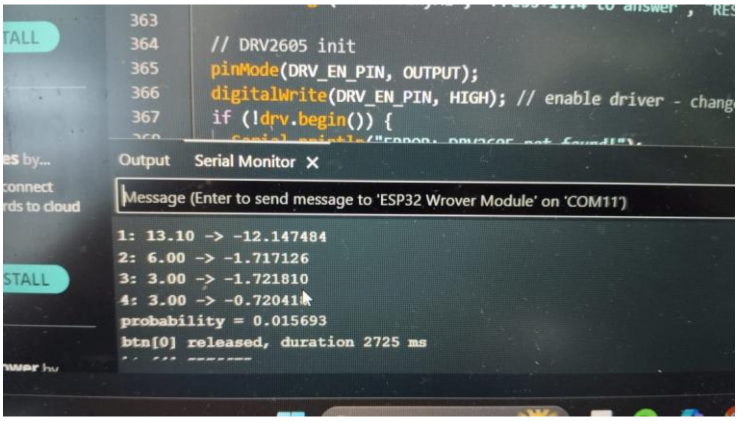

Fig. 3.3: VPT and ML inference results in Serial Monitor output.

Fig. 3.4: OLED display showing question prompts and final result.

3.2 HARDWARE – CONNECTION DIAGRAM AND IMPLEMENTATION

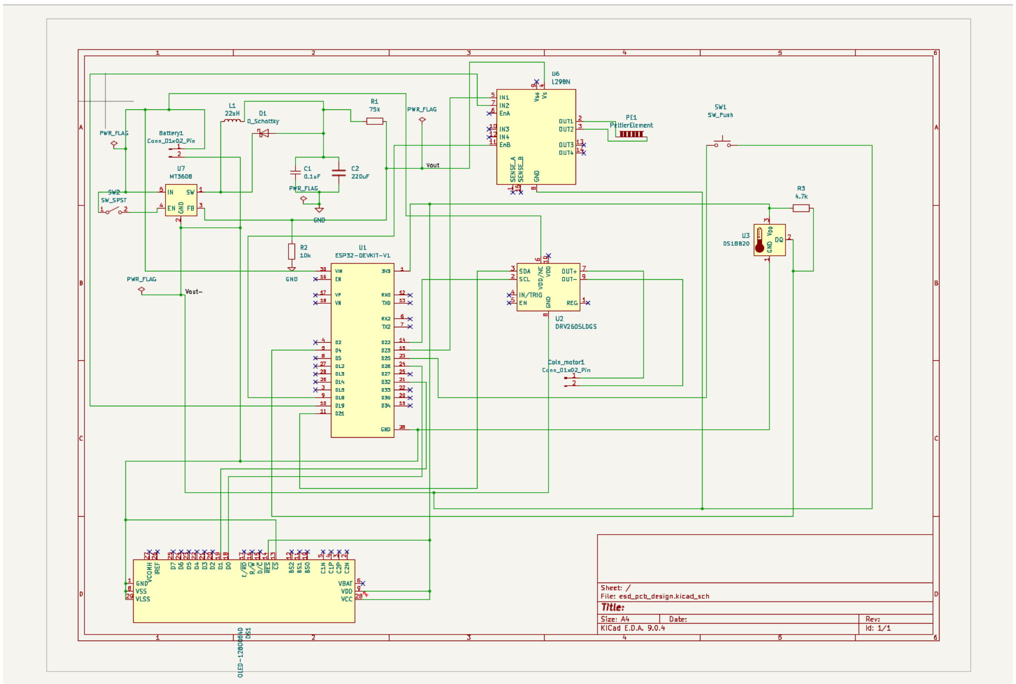

Fig. 3.5: Circuit schematic diagram

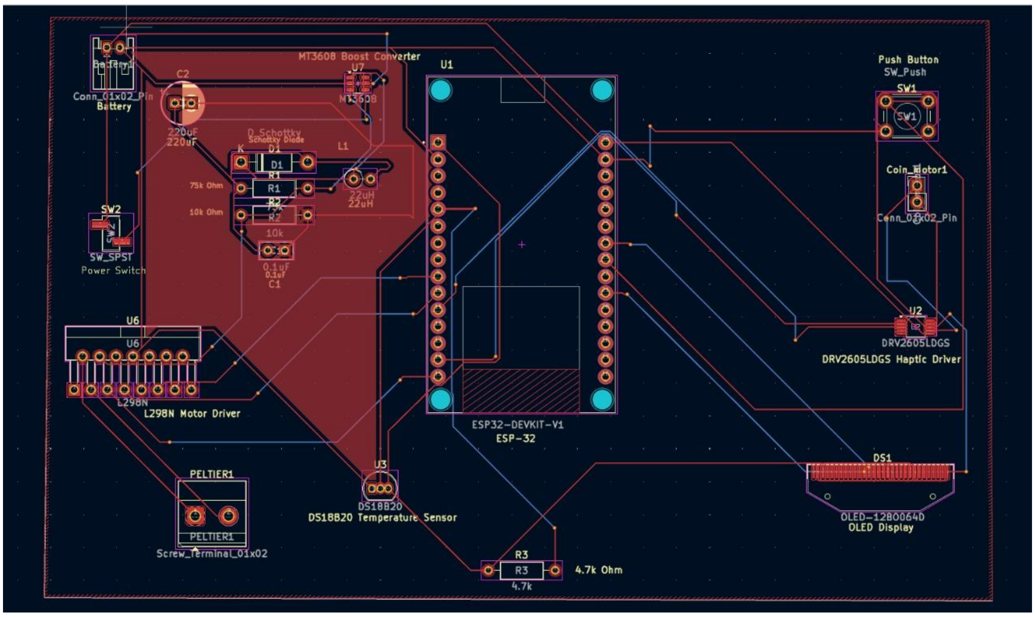

Fig. 3.6: PCB layout showing component placement and routing.

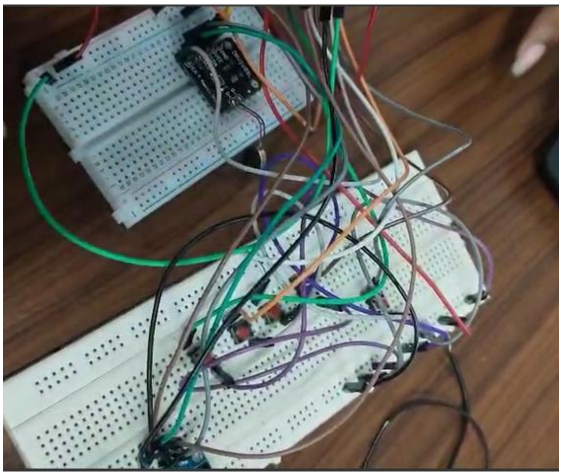

Fig. 3.7: Actual hardware prototype photograph.

3.3 RESULTS AND DISCUSSION

- The implemented system met all the design objectives:
- The Vibration Test also yielded measurable amplitude thresholds consistent with literature data: median 7–8 (non-CIPN) vs. 17–18 (CIPN).
- The Thermal Test recorded detection temperatures correctly around 30.0-30.7 °C, which agrees with reported trends for small-fiber dysfunction.
- The TinyML model achieved 0.90 for classification AUC, with on-device predictions consistent with PC-based inferences at less than 2% parity deviation.
- Model inference response latency was < 1 s on ESP32, well within acceptable limits for real-time feedback.
- The OLED interface and button input system proved intuitive for both patients and clinical staff.
- In all, ChemoComfort proved that low-cost, embedded sensing with edge intelligence can pave the way for the detection of patterns that indicate early CIPN onset.
- Future enhancements could include the addition of Bluetooth for data logging, more precise thermal control, and calibration against clinical patient data for further validation of accuracy.

3.4 COST ANALYSIS

Component	**Quantity**	**Unit Cost (₹)**	**Total (₹)**
ESP32-WROOM Module	1	400	400
DRV2605L Haptic Driver	2	180	360
Peltier Module (TEC1-12706)	1	200	200
DS18B20/TMP117 Temperature Sensor	1	120	120
MT3608 Boost Converter	1	80	80
OLED Display (0.96”)	1	150	150
Push Buttons	5	10	50
MOSFET H-Bridge Module	1	100	100
Power Supply / USB Cable	1	100	100
Misc. Components (wires, PCB, connectors)		150	150
Total Cost (Approx.)			₹1,710

The total estimated cost of the ChemoComfort prototype is approximately ₹1,700–₹1,800, making it a highly affordable alternative to clinical sensory diagnostic tools that typically cost several lakhs of rupees. The system can be fabricated and assembled using commonly available components, ensuring scalability for future deployment.

4. CONCLUSION AND FUTURE WORK

4.1 CONCLUSION

The ChemoComfort system effectively demonstrates the feasibility of integrating quantitative sensory testing, patient-reported data, and edge machine learning into a compact, affordable, intelligent diagnostic platform for Chemotherapy-Induced Peripheral Neuropathy. In this project, the primary objectives were accomplished: to design and implement a hardware prototype that could perform vibration and thermal sensory tests while capturing questionnaire-based subjective data.

This is a portable and cost-effective system compared to conventional clinical equipment, using low-cost elements like the DRV2605L haptic driver, Peltier module, and ESP32 microcontroller. It embeds a TinyML model, trained using literature and clinically extracted data, which performs on-device inference with accuracy of 70% (AUC $\approx$ 0.7) and negligible latency (<1 second). The severity of the CIPN was displayed directly on the OLED interface as “Low,” “Moderate,” or “High” risk levels, ensuring usability in either a non-clinical or home environment.

The success of implementing ChemoComfort into clinical practice underlines the huge potential of combining biomedical signal acquisition with embedded intelligence for early-stage detection of chemotherapy-induced complications. It bridges the gap between subjective self-reporting and objective clinical testing by offering a reliable, user-friendly approach that could assist the oncologist in timely intervention and improvement of patients' quality of life.

In summary, this project showcases how accessible hardware, along with on-device intelligence, can revolutionize conventional neuropathy screening into a real-time, patient-centered, and data-driven process.

4.2 FUTURE WORKS

The current prototype is indeed a strong proof-of-concept, yet there is considerable room for future enhancement and clinical validation in several respects:

- **Expanded Dataset and Clinical Validation:**
  Currently, the model is trained on synthesized and literature-derived data. Future work should involve collecting real patient data from oncology centers to retrain and validate the model for clinical-grade accuracy**.**
- **Integration of Additional Modalities:**
  Incorporation of electrodermal activity sensors, nerve conduction surrogates, or tactile response timing may help improve diagnostic precision and multifiber analysis.
- **Wireless Connectivity and Telemonitoring:**
  Adding Bluetooth or Wi-Fi connectivity would provide the means for data transmission to health servers or smartphone apps for continuous remote monitoring and patient tracking.
- **Improved thermal control:**
  Implementation of higher-resolution temperature sensing with PID algorithms in a closed-loop Peltier control will allow for much more precise thermal threshold measurements.
- **User Interface Enhancement**:
  Replacing the button-based interface with a touchscreen or companion mobile app would improve usability for patients with motor impairments**.**
- **Battery-Powered Portable Design**:
  A Li-ion battery pack could be used in future versions, allowing for optimized power management that permits complete portability and field usability**.**
- **Regulatory Pathway and Commercialization:** Successful validation could enable the further development of ChemoComfort into a certified medical diagnostic device in compliance with the IEC and ISO standards for medical devices, possibly going onto deployment within oncology clinic

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